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Practical 5

1)

Program :

```
import matplotlib.pyplot as plt

import pandas as pd


# Data for Months and Temperature for three cities

data = {

    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December'],

    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],

    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],

    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]

}


# Write your code...

df = pd.DataFrame(data)


plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_B_Temperature'],df['City_C_Temperature'])


plt.xlabel('Month')

plt.ylabel('Temperature')

plt.title('Temperature Variation')
```

```
plt.legend(loc='upper left')

plt.show()#
```

Output :

2)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Histogram

plt.hist(data['Age'],bins=30, edgecolor='k')


plt.title('Age Distribution')


plt.xlabel('Age')


plt.ylabel('Frequency')


plt.show()
```

Output :

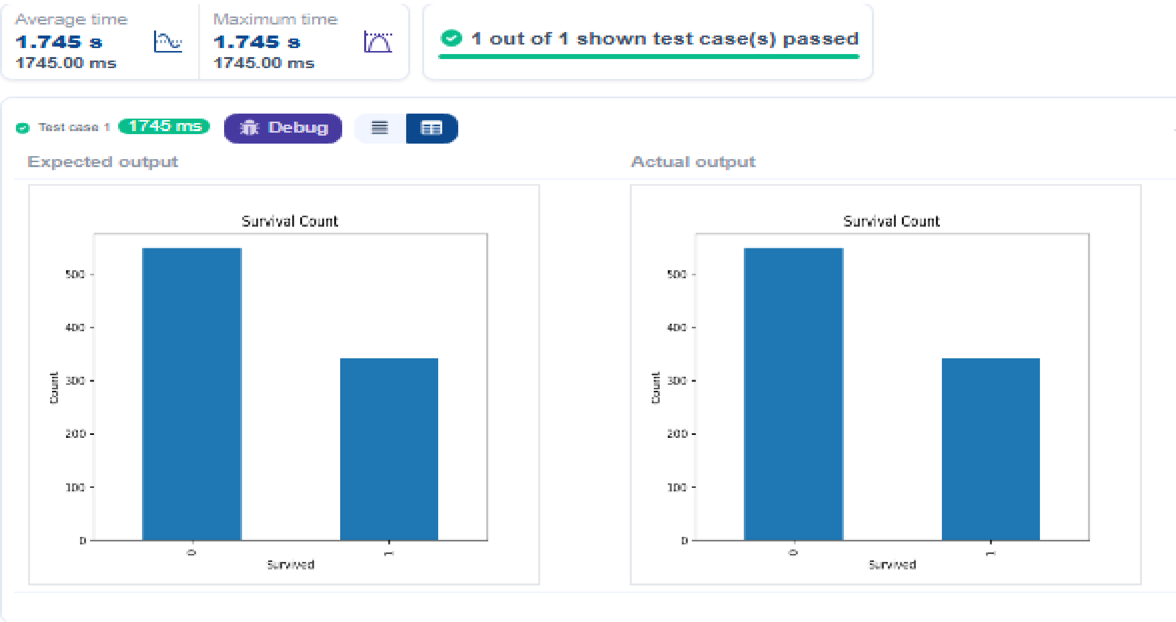
3)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)


# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Bar Plot for Survival Rate

survival_counts=data['Survived'].value_counts()


survival_counts.plot(kind='bar')


plt.title('Survival Count')


plt.xlabel('Survived')


plt.ylabel('Count')
```

```
plt.show()
```

Output :



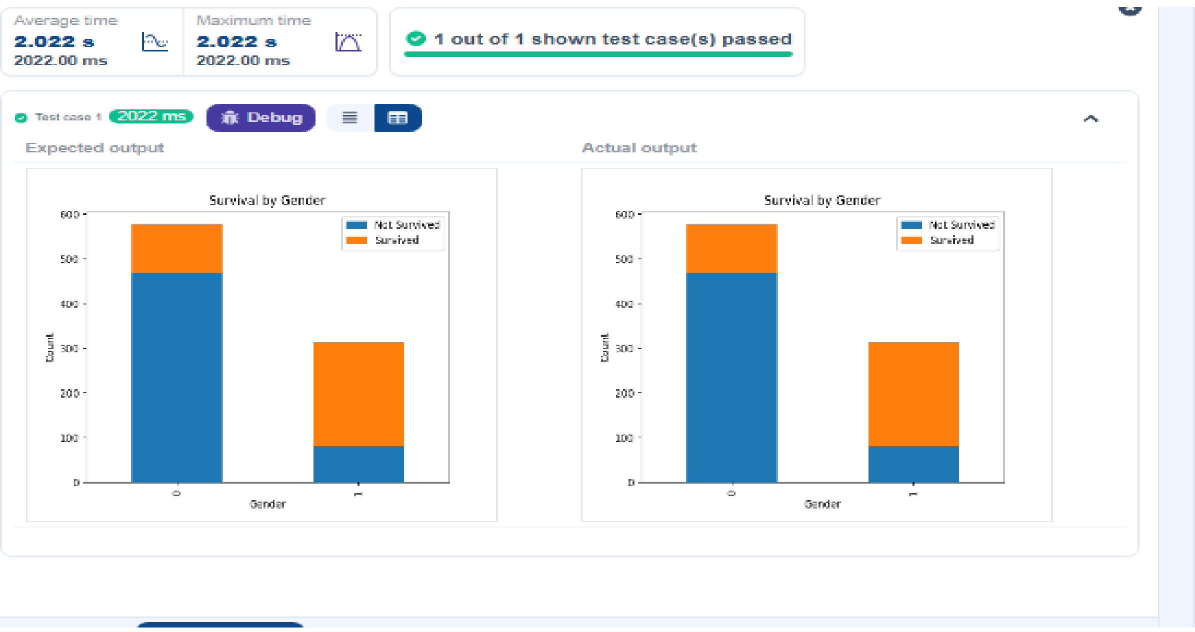
4)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Bar Plot for Survival by Gender

grouped=data.groupby('Sex')['Survived'].value_counts().unstack()

grouped.plot(kind='bar',stacked=True)

plt.title('Survival by Gender')

plt.xlabel('Gender')

plt.ylabel('Count')

plt.legend(['Not Survived','Survived'])

plt.show()
```

Output :

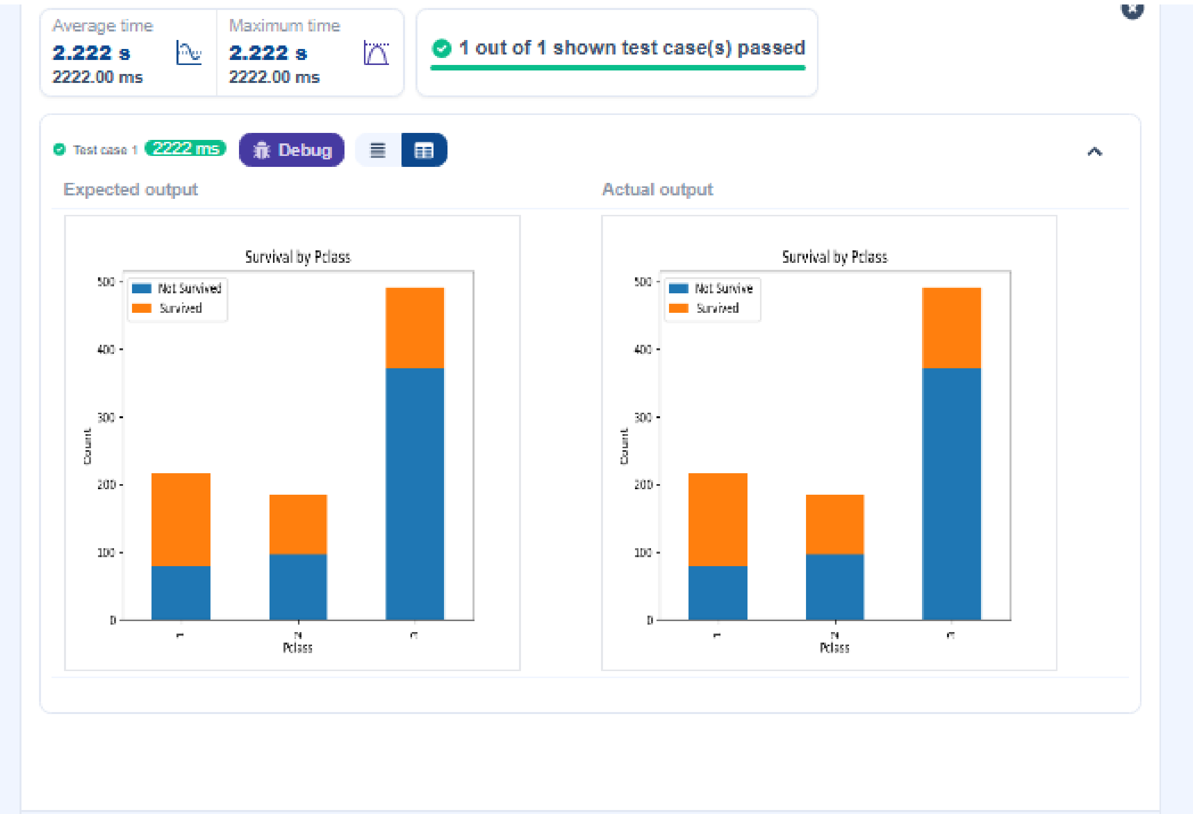


5)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)


# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Bar Plot for Survival by Pclass

grp=data.groupby("Pclass")["Survived"].value_counts().unstack()


grp.plot(kind="bar",stacked="True")

plt.xlabel("Pclass")


plt.ylabel("Count")


plt.title("Survival by Pclass")


plt.legend(["Not Survive ","Survived"])

plt.show()
```

Output :



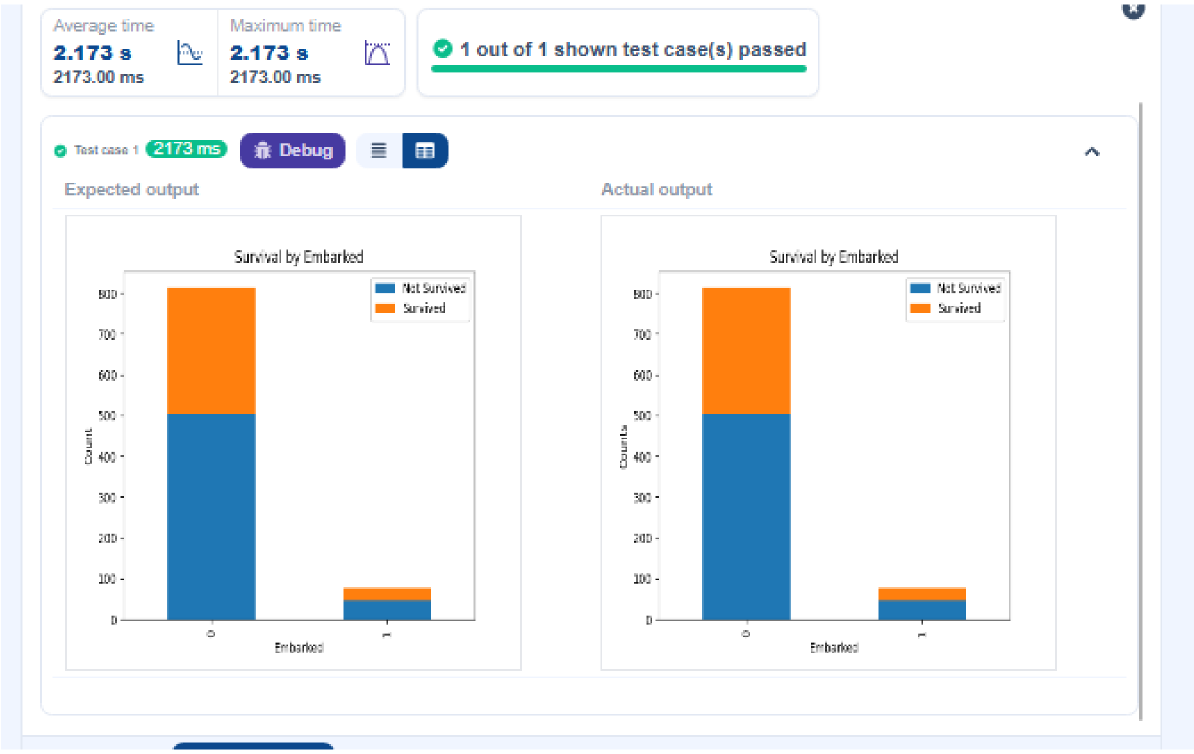
6)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Bar Plot for Survival by Embarked

grp=data.groupby("Embarked_Q")["Survived"].value_counts().unstack()


grp.plot(kind="bar",stacked="True")


plt.legend(["Not Survived", "Survived"])


plt.ylabel("Counts")


plt.xlabel("Embarked")


plt.title("Survival by Embarked")


plt.show()
```

Output :



7)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)


# Convert categorical features to numeric
```

```
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Box Plot for Age by Pclass


data.boxplot(column = 'Age', by='Pclass')

plt.title('Age by Pclass')

plt.suptitle(' ')

plt.xlabel('Pclass')

plt.ylabel('Age')

plt.show()

#
```

Output :

8)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)


# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Box Plot for Age by Survived


plt.figure(figsize=(8,6))

data.boxplot(column='Age',by='Survived')

plt.title('Age by Survival')

plt.suptitle(' ')

plt.xlabel('Survived')

plt.ylabel('Age')

plt.show()
```

Output :

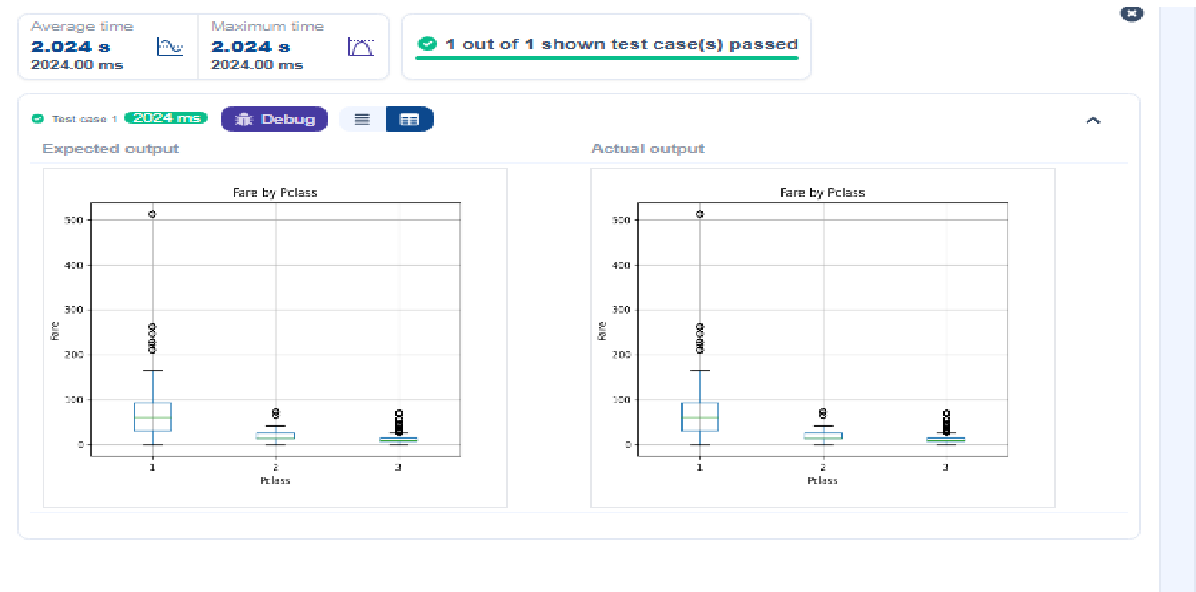
9)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Box Plot for Fare by Pclass


data.boxplot(column='Fare',by='Pclass')


plt.title('Fare by Pclass')

plt.suptitle('')

plt.xlabel('Pclass')

plt.ylabel('Fare')

plt.show()
```

Output :



10)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)


# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Box Plot for Fare by Pclass

plt.scatter(data['Age'],data['Fare'])


plt.title("Age vs. Fare")

plt.xlabel("Age")

plt.ylabel("Fare")

plt.show()
```

Output :

11)

Program :

```
import pandas as pd

import matplotlib.pyplot as plt


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')


# Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# Write your code here for Scatter Plot for Age vs. Fare by Survived


colors=data['Survived'].map({0: 'red',1: 'blue'})

plt.scatter(data['Age'],data['Fare'],c=colors)


plt.title("Age vs. Fare by Survival")

plt.xlabel('Age')

plt.ylabel('Fare')

plt.show()
```

Output :